

QUAV-Style Baseline Results on the Ibrahim UAV HUBO Scenario

Purpose and status

This report summarizes the uploaded QUAV-style baseline run. The implementation is a reproduction-style baseline inspired by the QUAV paper; it is not the official QUAV code. It is therefore appropriate as a same-platform reference baseline, but not as a claim that the official QUAV implementation was reproduced exactly.

Scenario: 8x8 grid, horizon $L=20$, start=(0, 0), target=(7, 7), obstacles=6. Candidate paths used by the QAOA selector: 12.

QAOA selector summary: selected candidate index 0; best one-hot state probability 0.4651; total runtime 5.14 s.

Result table

Method	Full success	Length	Turns	Buffer	Visibility	QUAV cost	HUBO energy	Runtime
A*	Yes	14	7	4	85.0%	37.50	-20742.26	0.028 s
RRT-best	Yes	14	5	4	70.0%	36.50	-20700.19	N/A
QUAV-style QAOA	Yes	14	1	3	80.0%	29.50	-20725.94	4.110 s

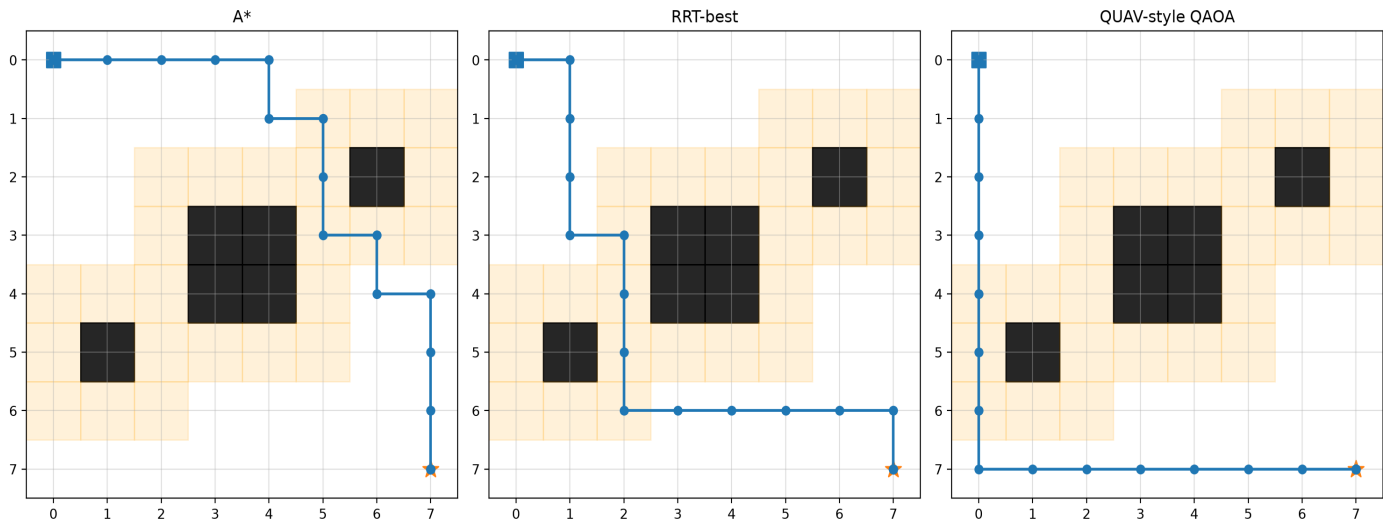
Interpretation

- All three methods return collision-free, valid-motion, target-reaching paths on this single benchmark instance. This means the QUAV-style baseline is usable as a same-platform comparison point.
- The QUAV-style QAOA selector chooses the lowest QUAV-style cost path: length 14, only one turn, three buffer steps, and 80.0% visibility. This is a clean geometric path, but it is not the lowest HUBO-energy path among the methods shown.
- A* has better visibility (85.0%) and lower HUBO energy than the QUAV-style selected path, although it has more turns and a higher QUAV-style cost. This shows that the QUAV-style objective and our HUBO objective are not identical.
- The comparison is fair as a platform-level baseline because the same map, start, target, obstacles, horizon, and feasibility checker are used. It is not fair to describe it as an exact comparison against official QUAV code because no official public implementation was available and this baseline selects among a limited set of generated candidate paths.

Recommended paper wording

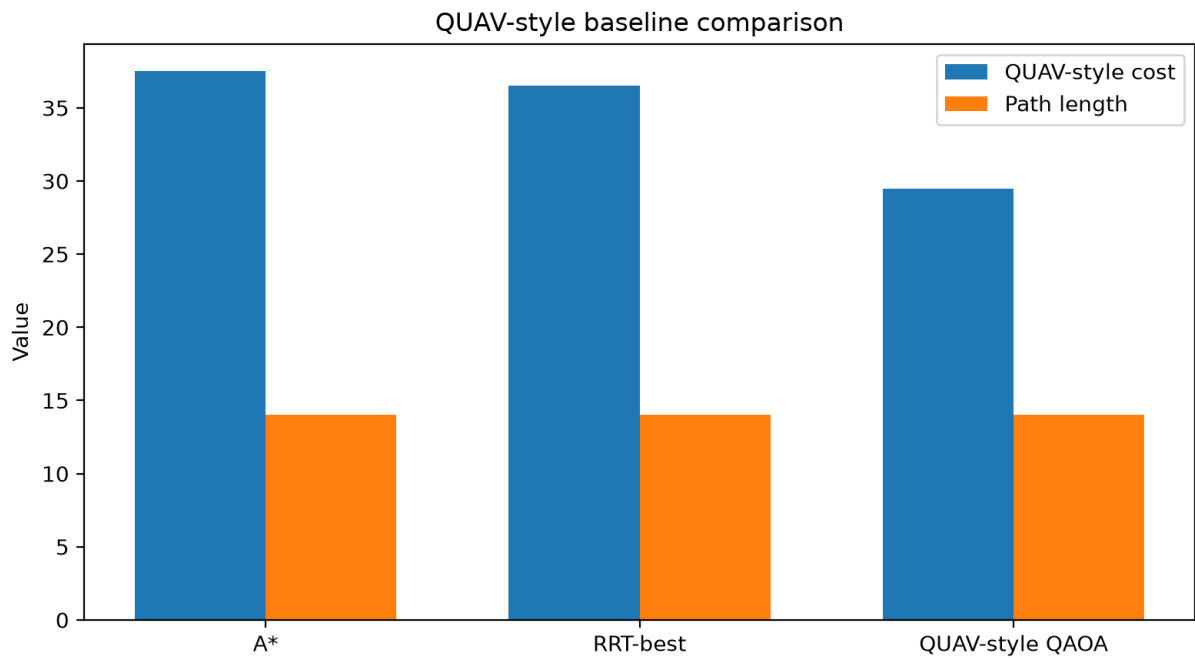
Because no public official QUAV implementation was available, we implemented a QUAV-style candidate-path QAOA baseline following the published algorithmic description. This baseline is used only as a same-platform reference and not as an exact reproduction of the authors' implementation. On our 8x8, $L=20$ benchmark, the QUAV-style selector produced a collision-free target-reaching path with lower QUAV-style geometric cost than A* and RRT-best, but A* retained higher visibility and lower native HUBO energy. These results support the distinction between path-geometry optimization and feasibility-aware HUBO/QUBO solver analysis.

Path comparison figure



The QUAU-style path follows the left and bottom boundary, giving a one-turn shortest path. A* takes a route with higher visibility. RRT-best is a valid exploratory path with lower visibility.

Metric figure



The QUAU-style selector has the lowest QUAV-style cost on this candidate set, while path length is identical for all three reported paths.